



Project Status report

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Community & UN SDG(s): **SaskTel network engineers and architects**

UN SDG(s):

- SDG#7: Affordable and clean energy
- SDG#11: Sustainable cities and communities
- SDG#12: Responsible consumption and production
- SDG#13: Climate action

MVP # **2 & 3**

Sprint cycle dates: **February 25th, 2025 - March 10th, 2025**

Project Name	Eco-Resilient Networks: Smart Deployment for the Future
Blurb	We are developing a simulation framework to optimize the deployment of Service Function Chains (SFCs) in edge-cloud environments for 5G video streaming, with a focus on <i>jointly</i> maximizing network availability and minimizing the carbon footprint. This project addresses UN Sustainable Development Goals 7, 9, 11, 12, and 13, aiming to provide network engineers and architects (specifically at SaskTel) with tools for data-driven decisions, leading to more efficient and sustainable network operations.
For Week Ending	March 10, 2025
Project Status	Green
Status Description	The project remains on track, and we have made significant progress on MVP 2. We have successfully implemented the three embedding policies (Availability-Aware, Carbon-Aware, and Tradeoff-Aware), integrated the Particle Swarm Optimization (PSO) algorithm, and implemented the calculation of key performance metrics (availability, carbon footprint, and latency). We have also begun initial comparisons of the different policies and have a working demo showcasing the optimization process. We are also making consistent progress on documentation (MVP 3).
Activities—During the past sprint cycle	

During the past sprint cycle we achieved the following:

- **Completed Work Package 2.1:** Embedding Policy Implementation.
 - Successfully implemented the Availability-Aware Embedding (AAE) policy.
 - Successfully implemented the Carbon-Aware Embedding (CAE) policy.
 - Successfully implemented the Tradeoff-Aware Embedding (TAE) policy.
- **Completed Work Package 2.2:** Redundancy Optimization Algorithm Implementation.
 - Successfully implemented the basic Particle Swarm Optimization (PSO) algorithm.
 - Successfully integrated PSO with the embedding policies.
- **Completed Work Package 2.3:** Performance Metric Calculation.
 - Implemented functions to calculate availability, carbon footprint, and latency.
- **Began Work Package 2.4:** Algorithm Comparison and Evaluation.
 - Designed an initial experiment matrix.
 - Ran preliminary simulations with different configurations.
- **Work Package 3.1:**
 - Code Documentation: Continued documenting the code using docstrings and comments.
- **Work Package 3.3:**



- Research Paper (Draft): Expanded the Methodology section to include further details on the implemented algorithms and policies.
- Started the performance evaluation section, incorporating initial system setup.
- Prepared presentation for Scrum #2.
- **Code/Demo:**
 - Demonstrate the simulation running with the TAE policy and PSO optimization.
 - Show how changing parameters (e.g., the weighting factor in TAE) affects the results.
 - Show the visualizations of network topology, VNF placement, and key performance metrics.
 - Briefly show the code for the embedding policies and PSO algorithm.

Project Issues

All identified minor issues have been successfully addressed and resolved. We maintain a proactive approach to issue management, ensuring no roadblocks impede our progress.

Project Changes

No project changes were required during this sprint cycle. The project scope and requirements remain well-defined and aligned with our initial plan.

Activities—Planned for Next Week

Following is the list of all activities that the team members will work on during the next week of the project:

- **Complete Work Package 2.4:** Algorithm Comparison and Evaluation (run more extensive simulations, analyze results, generate comparison reports).
- **Begin/continue Work Package 2.5:** Begin implementation of other algorithms (SA and GA).
- **Work Package 3.1: Code Documentation:** Continue documenting new code as it is developed.
- **Work Package 3.2: User Guide:** Begin drafting the user guide, focusing on how to configure and run the simulation.
- **Work Package 3.3: Research Paper (Draft):** Continue writing the Results section, incorporating more comprehensive findings. Begin outlining the Discussion and Conclusion sections.

Reflection

Do you feel "on track"?

Yes, the project is currently on track according to the planned schedule.

What progress do you particularly feel good (great) about?

We are particularly pleased with the successful integration of the PSO algorithm with the embedding policies and the ability to visualize the optimization process. The initial results demonstrating the tradeoff between availability and sustainability are very encouraging.

What barriers (if any) do you feel is/are a current impediment to success?

No significant barriers have been encountered so far.

What help (if any) do you require to move positively forward?

We welcome feedback on the design of our experiments and the clarity of our visualizations.

What questions or concerns do you have (if any)?

We are currently focusing on PSO. We are interested in feedback on the relative importance of implementing SA and GA within the remaining project timeframe.